



LOKMANYA TILAK JANKALYAN SHIKSHAN SANSTHA'S
PRIYADARSHINI COLLEGE OF ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

HACKATHON 2026

Idea Submission Template | Code. Create. Innovate.

Team Name:

Team Raksha

Problem Statement No:

2

Problem Statement Title:

Improving Safety at Large Public Events

Team Leader Name:

Shounak Joshi

IDEA TITLE

Know the crowd. Know the person. Stop the surge.

Cameras watch. The software remembers faces.
It warns guards before a crush builds.

0.4s

face recognised

<1s

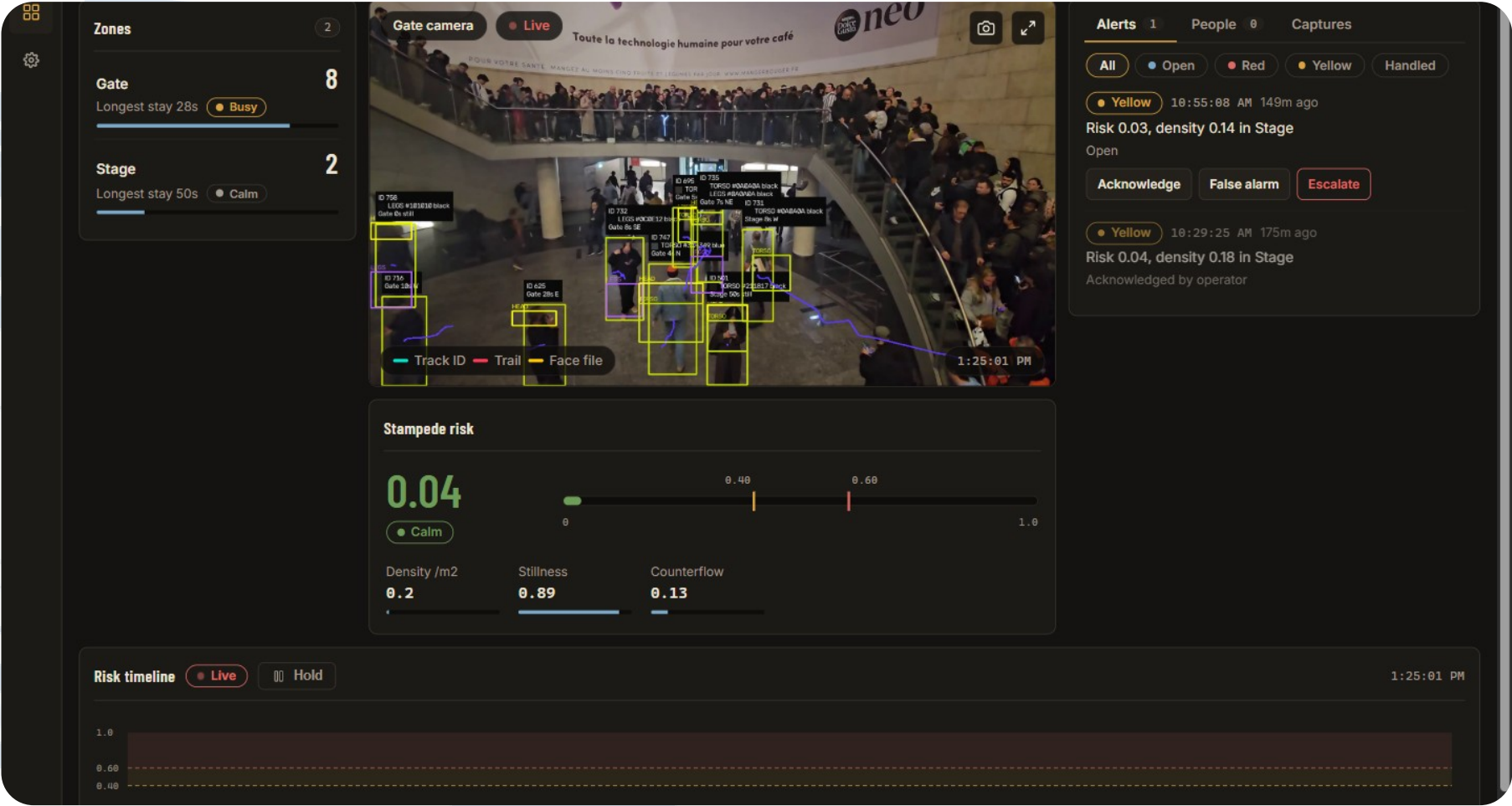
guard warned

15

video checks every second

Watch it work

- 1 A face appears on camera.
- 2 Their file pops up in 0.4 seconds.
- 3 A crowded zone turns yellow in under 1 second.

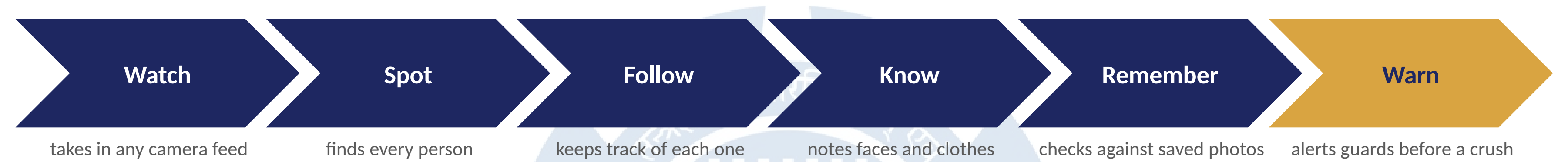


The live wall, running on real station footage.

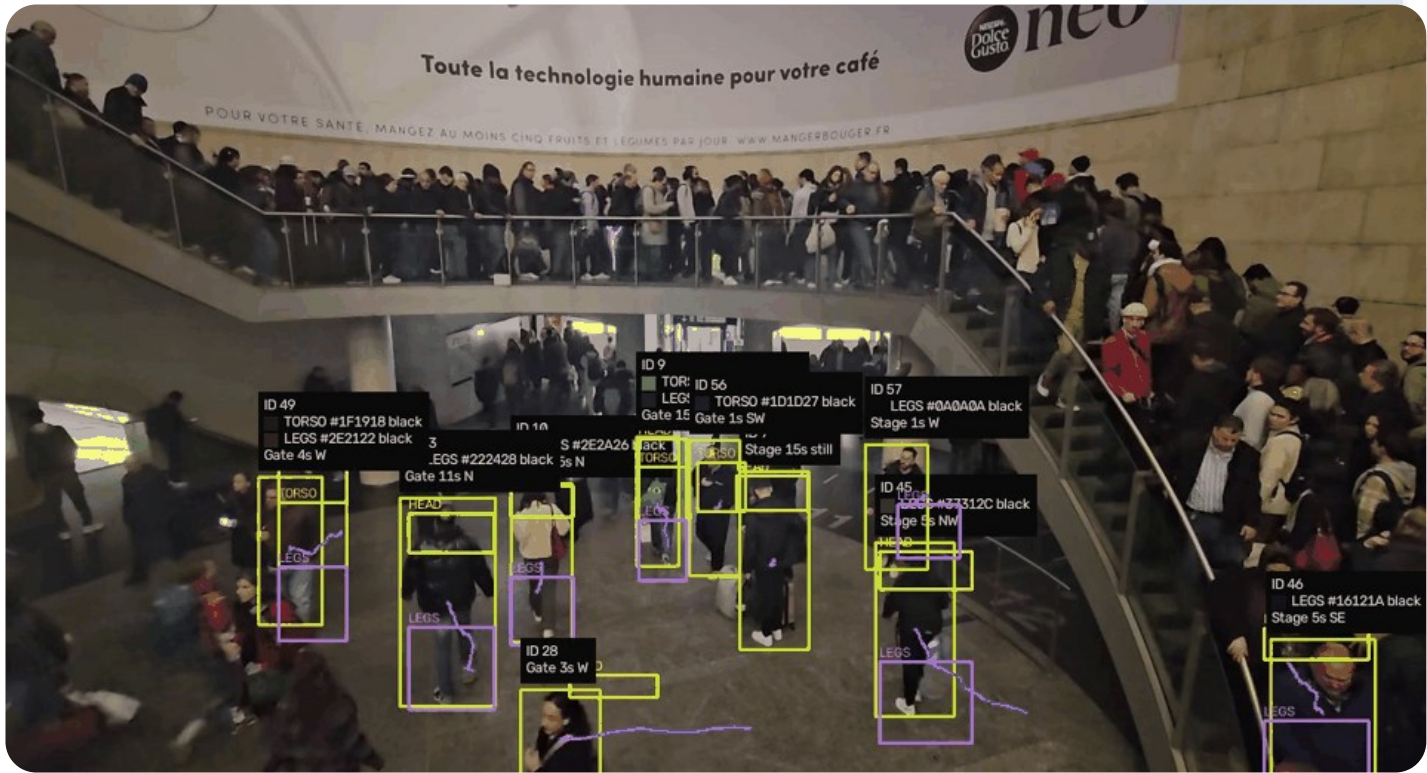
What the colours mean

- **Calm**
All is well.
- **Busy**
Getting crowded. Guards get an early warning.
- **Danger**
Crowd packed and stuck. Clear the area now.

TECHNICAL APPROACH



In plain words: the camera watches, the software follows every person, and it warns guards before a crush builds.



Keeps only good photos

Only sharp, front-facing photos are saved. Blurry ones are ignored, so matches stay reliable.

Tested on real video

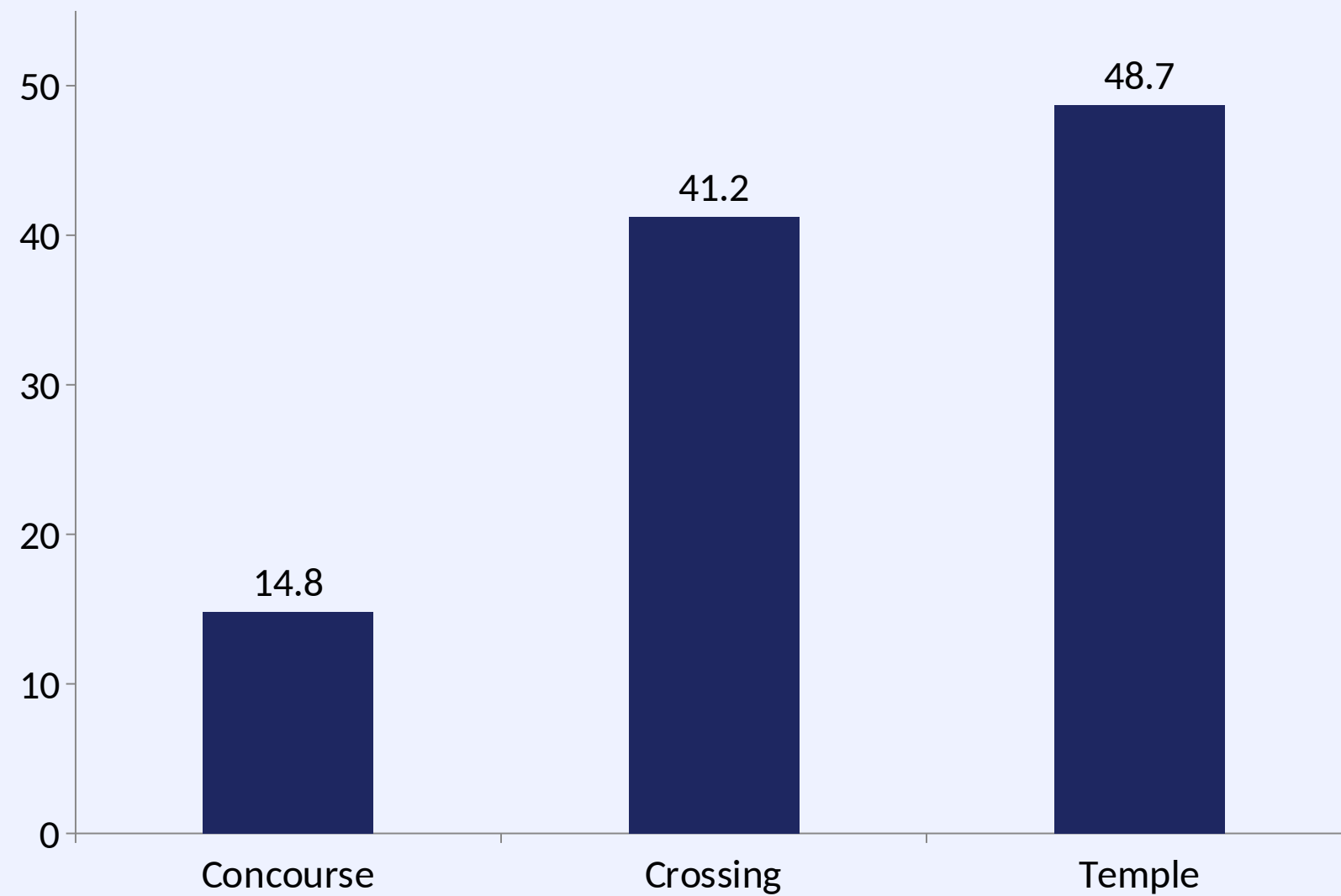
Followed 18 people at once and kept track of 850 people across 2 minutes of real station video.

Works with normal cameras

Any ordinary camera feed can be plugged in. No special equipment needed.

FEASIBILITY AND VIABILITY

Video checks per second (higher is better)



Tested on real crowd video from three places, all on one gaming laptop. Higher means smoother.

Instant

A face is matched in just 0.06 seconds.

Red alert

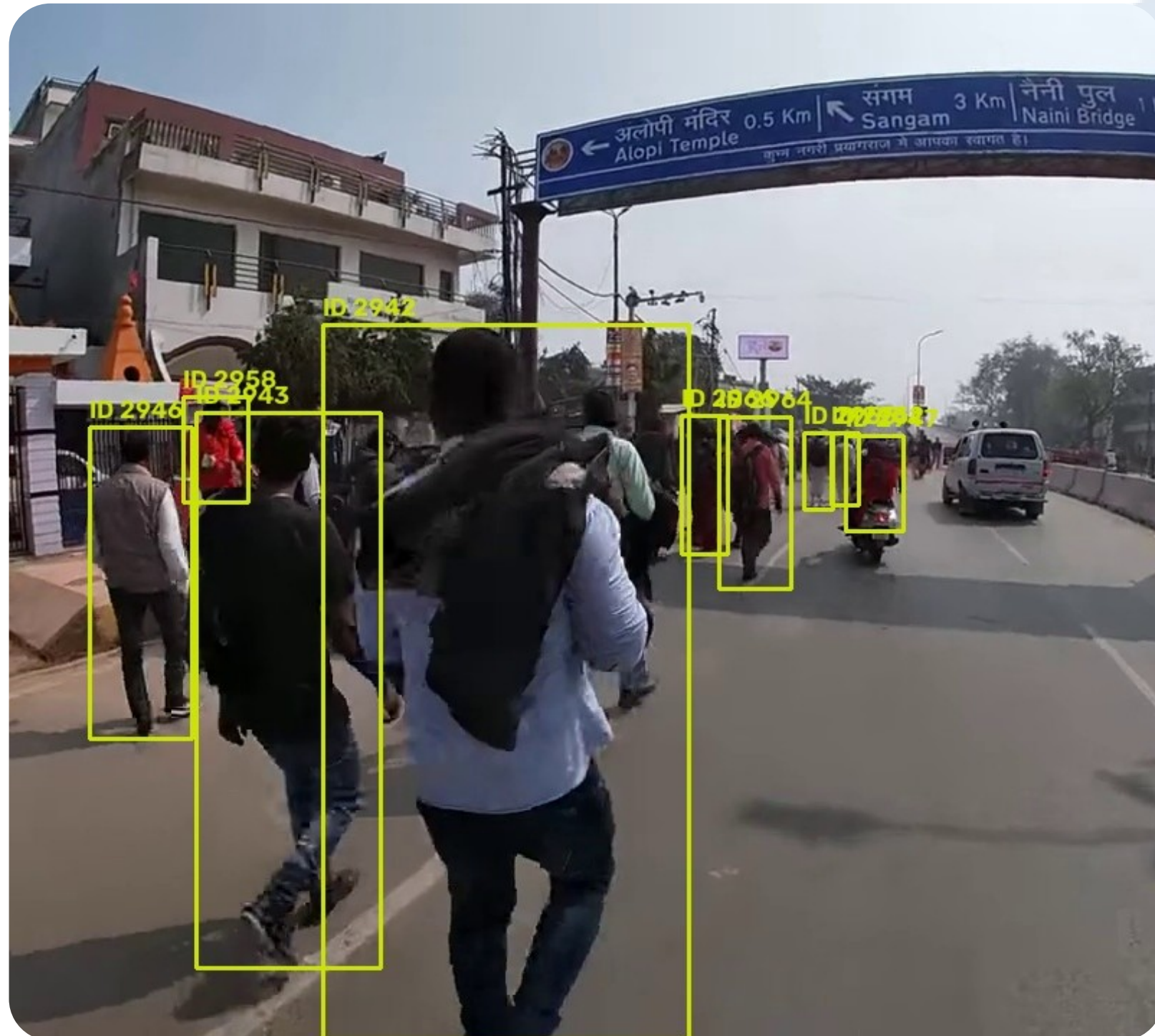
Fires when a crowd packs tight and stops moving.

Runs on one laptop

Even the heaviest crowd runs smoothly.



IMPACT AND BENEFITS



Before

50 screens, one tired guard, and risks get missed.

After

One screen watches all cameras and warns only about risky zones.

Organizers

Spot a growing jam about 60 seconds before people notice, and tell guards which zone to clear.

Police

Find a lost child or a suspect from a photo in under 2 seconds. Full history in one click.

Public

Safer exits and faster help. Faces are blurred when saved.

RESEARCH AND REFERENCES

The science behind it

Finding and following people

Finds every person and follows each one. Uses YOLOv8 and ByteTrack, rated in 2025-26 studies as the best mix of speed and accuracy.

Matching faces

Compares a face with saved photos. Uses ArcFace and Faiss, which score 99.5 percent on the standard face test.

Predicting crushes

Crowd science models (AURORA and FTLE) spot danger early. A Delhi trial cut response time by 37 percent.

Why India needs it

Hathras, Kumbh and railway station crushes all came from packed halls and blocked exits.

Privacy built in

Blurred storage

Faces are blurred in everything we save.

7-day auto delete

Old data is wiped after 7 days. Nothing piles up.

Tamper-proof log

Every action is recorded in a chain that cannot be quietly changed.

No ID linking

Only people who choose to opt in can be matched.